29.2: Mosses and other nonvascular plants have life cycles dominated by gametophytes

An evolutionary tree diagram highlights nonvascular plants (bryophytes).

Branch point 1: First level-1 branch: Nonvascular plants (bryophytes). The second level-1 branch splits from branch point 2 into the first level-2 branch: seedless vascular plants. The second level-2 branch splits from branch point 3 into the first level-3 branch: Gymnosperms and the second level-3 branch: Angiosperms.

The nonvascular plants (bryophytes) are represented today by three phyla of small, herbaceous (nonwoody) plants: **liverworts** (phylum Hepatophyta), **mosses** (phylum Bryophyta), and **hornworts** (phylum Anthocerophyta). Liverworts and hornworts are named for their shapes, plus the suffix *wort* (from the Anglo-Saxon for "herb"). Mosses are familiar to many people, although some plants commonly called "mosses" are not really mosses at all. These include Irish moss (a red seaweed), reindeer moss (a lichen), club mosses (seedless vascular plants), and Spanish mosses (lichens in some regions and flowering plants in others).

Phylogenetic analyses indicate that liverworts, mosses, and hornworts diverged from other plant lineages early in the history of plant evolution (see Figure 29.10). Fossil evidence provides some support for this idea: The earliest spores of plants (dating from 450 to 470 million years ago) have structural features found only in the spores of liverworts, and by 430 million years ago spores similar to those of mosses and hornworts also occur in the fossil record. The earliest fossils of vascular plants date to about 425 million years ago.

Over the long course of their evolution, liverworts, mosses, and hornworts have acquired many unique adaptations. Next, we'll examine some of those features.

Bryophyte Gametophytes

Unlike vascular plants, in all three bryophyte phyla the haploid gametophytes are the dominant stage of the life cycle: They are usually larger and longer-living than the sporophytes, as shown in the moss life cycle in **Figure 29.11**. The sporophytes are typically present only part of the time.

Figure 29.11

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The life cycle of a moss.
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Life cycle of a moss showing the male and female gametophyte.

The cycle is as follows.

1.Spores develop into threadlike protonemata (n).

2.The haploid protonemata produce buds that divide by mitosis and grow into gametophytes. Male gametophyte (n) have antheridia that produce many sperm. Female gametophyte (n) have archegonia that each produce one egg.

3.Sperm must swim through a film of moisture to reach the egg and then fertilization occurs with the archegonium. The Zygote (2 n) forms in the archegonium and

4.develops into a sporophyte embryo.

5.The sporophyte grows a long stalk (seta) that emerges from the archegonium. A capsule (sporangium) forms at the end of the seta.

6.Attached by its foot, the young sporophyte (2 n) remains nutritionally dependent on the gametophyte.

7.Meiosis occurs and haploid spores develop in the capsule. When the capsule is mature, its lid pops off, and the spores are released through the peristome.

VISUAL SKILLS *In this diagram, does the sperm cell that fertilizes the egg cell differ genetically from the egg? Explain.*

*For suggested answer, see*Appendix A.**

Animation

Moss Life Cycle

image

Play

image

Play

When bryophyte spores are dispersed to a favorable habitat, such as moist soil or tree bark, they may germinate and grow into gametophytes. Germinating moss spores, for example, characteristically produce a mass of green, branched, one-cell-thick filaments known as a **protonema** (plural, *protonemata*). A protonema has a large surface area that enhances absorption of water and minerals. In favorable conditions, a protonema produces one or more “buds,” each of which develops into a moss gametophyte. (Note that when referring to nonvascular plants, we often use quotation marks for structures similar to the buds, stems, and leaves of vascular plants because the definitions of these terms are based on vascular plant organs.)

Bryophyte gametophytes generally form ground-hugging carpets, partly because their body parts are too thin to support a tall plant. A second constraint on the height of many bryophytes is the absence of vascular tissue, which would enable long-distance transport of water and nutrients. (The thin structure of bryophyte organs makes it possible to distribute materials for short distances without specialized vascular tissue.) However, some mosses have conducting tissues in the center of their “stems.” A few of these mosses can grow as tall as 60cm 2 feet as a result. Phylogenetic analyses suggest that conducting tissues similar to those of vascular plants arose independently in these mosses by convergent evolution.

The gametophytes are anchored to the ground by delicate **rhizoids**, which are long, tubular single cells (in liverworts and hornworts) or filaments of cells (in mosses). Unlike roots, which are found in vascular plant sporophytes, rhizoids lack specialized conducting cells and do not play a primary role in water and mineral absorption.

Gametophytes can form multiple****__gametangia__, multicellular structures that produce gametes and are covered by protective tissue. Female gametangia are called **archegonia**, and male gametangia are called **antheridia**. Each archegonium produces one egg, whereas each antheridium produces many sperm. Some bryophyte gametophytes are bisexual, but in mosses the archegonia and antheridia are typically carried on separate female and male gametophytes. Flagellated sperm swim through a film of water toward eggs, entering the archegonia in response to chemical attractants. Eggs are not released but instead remain within the bases of archegonia. After fertilization, embryos are retained within the archegonia. Layers of placental transfer cells help transport nutrients to the embryos as they develop into sporophytes.

Bryophyte sperm typically require a film of water to reach the eggs. Given this requirement, it is not surprising that many bryophyte species are found in moist habitats. The fact that sperm swim through water to reach the egg also means that in species with separate male and female gametophytes (most species of mosses), sexual reproduction is likely to be more successful when individuals are located close to one another.

Many bryophyte species can increase the number of individuals in a local area through various methods of asexual reproduction. For example, some mosses reproduce asexually by forming *brood bodies*, small plantlets that detach from the parent plant and grow into new, genetically identical copies of their parent **(Figure 29.12)**.

Figure 29.12

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A moss brood body.
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A moss brood body has a stem-like body with a tree-like top.

Bryophyte Sporophytes

The cells of bryophyte sporophytes contain plastids that are usually green and photosynthetic when the sporophytes are young. Even so, bryophyte sporophytes cannot live independently. A bryophyte sporophyte remains attached to its parental gametophyte throughout the sporophyte's lifetime, dependent on the gametophyte for supplies of sugars, amino acids, minerals, and water.

Bryophytes have the smallest sporophytes of all extant plant groups. A typical bryophyte sporophyte consists of a foot, a seta, and a sporangium. Embedded in the archegonium, the **foot** absorbs nutrients from the gametophyte. The **seta** (plural, *setae*), or stalk, conducts these materials to the sporangium, also called a **capsule**, which produces spores by meiosis.

Bryophyte sporophytes can produce enormous numbers of spores. A single moss capsule, for example, can generate over 5 million spores. In most mosses, the seta becomes elongated, enhancing spore dispersal by elevating the capsule. Typically, the upper part of the capsule features a ring of interlocking, tooth-like structures known as the **peristome** (see Figure 29.11). These "teeth" open under dry conditions and close again when it is moist. This allows moss spores to be discharged gradually, via periodic gusts of wind that can carry them long distances.

Moss and hornwort sporophytes are often larger and more complex than those of liverworts. For example, hornwort sporophytes, which superficially resemble grass blades, have a cuticle. Moss and hornwort sporophytes also have stomata, as do all vascular plants (but not liverworts).

Figure 29.13 shows some examples of gametophytes and sporophytes in the bryophyte phyla.

Figure 29.13

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Exploring Bryophyte Diversity
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Exploring bryophyte diversity.

Three plants are described as follows.

Liverworts (phylum Hepatophyta). This phylum's common and scientific names (from the Latin hepaticus, liver) refer to the liver-shaped gametophytes of its members, such as Marchantia, shown in a. In medieval times, their shape was thought to be a sign that the plants could help treat liver diseases. Some liverworts, including Marchantia, are described as thalloid because of the flattened shape of their gametophytes. Marchantia gametangia are elevated on gametophores that look like miniature trees. You would need a magnifying glass to see the sporophytes, which have a short seta (stalk) with an oval or round capsule. Other

liverworts, such as Plagiochila, shown in b, are called leafy because their stem-like gametophytes have many leaflike appendages. There are many more species of leafy liverworts than thalloid liverworts.

Hornworts (phylum Anthocerophyta). This phylum's common and scientific names (from the Greek keras, horn) refer to the long, tapered shape of the sporophyte. A typical sporophyte can grow to about 5 cm high. Unlike a liverwort or moss sporophyte, a hornwort sporophyte lacks a seta and consists only of a sporangium. The sporangium releases mature spores by splitting open, starting at the tip of the horn. The gametophytes, which are usually 1–2 centimeters in diameter, grow mostly horizontally and often have multiple sporophytes attached. Hornworts are frequently among the first species to colonize open areas with moist soils; a symbiotic relationship with nitrogen-fixing cyanobacteria contributes to their ability to do this (nitrogen is often in short supply in such areas).

The Ecological and Economic Importance of Mosses

Wind dispersal of lightweight spores has distributed mosses throughout the world. These plants are particularly common and diverse in moist forests and **wetlands**. Some mosses colonize bare, sandy soil, where, researchers have found, they help retain nitrogen in the soil **(Figure 29.14)**. In northern coniferous forests, species such as the feather moss *Pleurozium* harbor nitrogen-fixing cyanobacteria that increase the availability of nitrogen in the ecosystem. Other mosses inhabit such extreme environments as mountaintops, **tundra**, and deserts. Many mosses are able to live in very cold or dry habitats because they can survive the loss of most of their body water, then rehydrate when moisture is available. Few vascular plants can survive the same degree of desiccation. Moreover, phenolic compounds in moss cell walls absorb damaging levels of UV radiation present in deserts or at high altitudes.

Figure 29.14

Inquiry: Can bryophytes reduce the rate at which key nutrients are lost from soils?

Experiment

Soils in terrestrial ecosystems are often low in nitrogen, a nutrient required for normal plant growth. Richard Bowden, of Allegheny College, measured annual inputs (gains) and outputs (losses) of nitrogen in a sandy soil ecosystem dominated by the moss *Polytrichum*. Nitrogen inputs were measured from rainfall (dissolved ions, such as nitrate, NO_3^-), biological N_2 fixation, and wind deposition. Nitrogen losses were

measured in leached water (dissolved ions, such as NO_3^-) and gaseous emissions (such as N_2O emitted by bacteria). Bowden measured losses for soils with *Polytrichum* and for soils where the moss was removed two months before the experiment began.

Results

A total of 10.5kg of nitrogen per hectare (kg/ha) entered the ecosystem each year. Little nitrogen was lost by gaseous emissions 0.10 kg/ha·yr. The results of comparing nitrogen losses by leaching are shown below.

A bar graph plots annual nitrogen loss versus moss. A bar for, with moss, rises to approximately 0.40, and the bar for, without moss, rises to approximately 5.2.

Conclusion

The moss *Polytrichum* greatly reduced the loss of nitrogen by leaching in this ecosystem. Each year, the moss ecosystem retained over 95% of the 10.5kg/ha of total nitrogen inputs (only 0.1kg/ha and 0.3kg/ha were lost to gaseous emissions and leaching, respectively).

Data from R. D. Bowden, Inputs, outputs, and accumulation of nitrogen in an early successional moss (*Polytrichum*) ecosystem, *Ecological Monographs* 61:207–223 (1991).

WHAT IF?*How might the presence of* Polytrichum *affect plant species that typically colonize the sandy soils after the moss?*

*For suggested answer, see*Appendix A*.*

One wetland moss genus, *Sphagnum*, or peat moss, is often a major component of deposits of partially decayed organic material known as **peat(Figure 29.15a)**. Boggy regions with thick layers of peat are called peatlands. *Sphagnum* does not decay readily, in part because of phenolic compounds embedded in its cell walls. The low temperature, pH, and oxygen level of peatlands also inhibit decay of moss and other organisms in these boggy wetlands. As a result, some peatlands have preserved corpses for thousands of years **(Figure 29.15b)**.

Figure 29.15

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    *Sphagnum*, or peat moss: a bryophyte with economic,
ecological, and archaeological significance.
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Part a. Peat being harvested from a peatland. Part b. Tollund Man, a bog mummy dating from 405-100 B.C.E. The acidic, oxygen-poor conditions produced by Sphagnum can preserve human or other animal bodies for thousands of years.

Peat has long been a fuel source in Europe and Asia, and it is still harvested for fuel today, notably in Ireland and Canada. Peat moss is also useful as a soil conditioner and for packing plant roots during shipment because it has large dead cells that can absorb roughly 20 times the moss's weight in water.

Peatlands cover just 3% of Earth's land surface but contain one-third of the world's soil carbon: Globally, an estimated 500 billion tons of organic carbon is stored as peat. Current overharvesting of *Sphagnum*—primarily for use in peat-fired power stations—contributes to global warming by releasing stored CO_2. In addition, if global temperatures continue to rise, the water levels of some peatlands are expected to drop. Such a change would expose peat to air and cause it to decompose, thereby releasing additional stored CO_2 and contributing further to global warming. The historical and expected future effects of *Sphagnum* on the global climate underscore the importance of preserving and managing peatlands.

Mosses may have a long history of affecting climate change. In the **Scientific Skills Exercise**, you will explore the question of whether they did so during the Ordovician period by contributing to the weathering of rocks.

Scientific Skills Exercise

Making Bar Graphs and Interpreting Data

AP Science Practices 1, 3, 4

Could Nonvascular Plants Have Caused Weathering of Rocks and Contributed to Climate Change During the Ordovician Period?The oldest traces of terrestrial plants are fossilized spores formed 470 million years ago. Between that time and the end of the Ordovician period 444 million years ago, the atmospheric CO_2 level dropped by half, and the climate cooled dramatically.

One possible cause of the drop in CO_2 during the Ordovician period is the breakdown, or weathering, of rock. As rock weathers, calcium silicate (Ca_2SiCO_3) is released and combines with CO_2 from the air, producing calcium carbonate ($CaCO_3$). Today, the roots of vascular plants increase rock weathering and mineral release by producing acids that break down rock and soil. Although nonvascular plants lack roots, they require the same mineral nutrients as vascular plants. Could nonvascular plants also increase the chemical weathering of rock? If so, they could have contributed to the decline in atmospheric CO_2 during the

Ordovician. In this exercise, you will interpret data from a study of the effects of moss on releasing minerals from two types of rock.

Fossilized spores.

How the Experiment Was Done The researchers set up experimental and control microcosms, or small artificial ecosystems, to measure mineral release from rocks. First, they placed rock fragments of volcanic origin, either granite or andesite, into small glass containers. Then they mixed water and macerated (chopped and crushed) moss of the species *Physcomitrella patens*. They added this mixture to the experimental microcosms (72 granite and 41 andesite). For the control microcosms (77 granite and 37 andesite), they filtered out the moss and just added the water. After 130 days, they measured the amounts of various minerals found in the water in the control microcosms and in the water and moss in the experimental microcosms.

Data from the Experiment The moss grew (increased its biomass) in the experimental microcosms. The table shows the mean amounts in micromoles (µmol) of several minerals measured in the water and the moss in the microcosms.

A table illustrates values of granite and andesite for calcium, magnesium, and potassium ions.

Data from T. M. Lenton et al., First plants cooled the Ordovician, *Nature Geoscience* 5:86–89 (2012).

	C a superscript 2 plus (micromoles)	**M g superscript 2 plus (micromoles)**	**K superscript plus (micromoles)**
Granite	Andesite	Granite	Andesite
Mean weathered amount released in water in the control microcosms	1.68	1.54	0.42
Mean weathered amount released in water in the experimental microcosms	1.27	1.84	0.34
Mean weathered amount taken up by moss in the experimental microcosms	1.09	3.62	0.31

INTERPRET THE DATA

- Why did the researchers add filtrate from which macerated moss had been removed to the control microcosms?

- Make two bar graphs (for granite and andesite) comparing the mean amounts of each element weathered from rocks in the control and experimental microcosms. (Hint: For an experimental microcosm, what sum represents the total amount weathered from rocks?) (For additional information on bar graphs, see the Scientific Skills Review in Appendix D.)
- Overall, what is the effect of moss on chemical weathering of rock? Are the results similar or different for granite and andesite?
- Based on their experimental results, the researchers added weathering of rock by nonvascular plants to simulation models of the Ordovician climate. The new models predicted decreased CO_2 levels and global cooling sufficient to produce the glaciations in the late Ordovician period. What assumptions did the researchers make in using results from their experiments in climate simulation models?
- "Life has profoundly changed the Earth." Explain whether or not these experimental results support this statement.

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.